

Rishabh Singh Thakur

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Summary

Helping businesses leverage AI and cloud infrastructure to automate workflows, reduce operational costs, and accelerate time-to-market. AI Cloud Engineer & SRE with around 5 years of hands-on experience building scalable GCP infrastructure with HL7 FHIR compliance for 1000+ SDM stores across Canada. Built 3 full stack AI SaaS apps with one production grade app with google auth, stripe payments and OWASP security integration.

Work experience

Senior Site Reliability Engineer

Loblaw Companies Limited

TORONTO, CANADA

03/2025 – Present

K8s Store Platform Rollout for SDM Anthos and Vanilla Kubernetes

- Engineered **Cursor** agent workflows, skills, and rules to generate **GitLab** pipelines and **Ansible** automation moving SDM from on-prem **VMware** to **vanilla Kubernetes** on **GCP** (~400 stores this year, 1,500+ footprint), including node rebuilds, container/**KubeVirt** migration, and **FluxCD** **GitOps**.

AI Engineered DevOps Delivery for Pharmacy Platform Migrations

- Built a **Cursor** AI delivery system (**Agent Skills, Rules, MCP**) that orchestrates tickets, repo changes, docs, and **GitLab** jobs across pharmacy SRE repos serving 1,500+ stores; applied it to **DB2** → **Postgres** (convert/cutover/rollback pipelines, pilots underway) and **IBM MQ** → **ActiveMQ** (side-by-side brokers, dump-driven store-by-store migrate).

AI Engineered K8s Incident Intelligence and Store Recovery

- Built an AI incident system in **Cursor** over 10,000+ store incidents (1,500+ estate): RCA playbooks, K8s store-down recovery runbooks, and the first **GitLab** recovery jobs; during an ~80-store release, traced **GitLab/Pub/Sub/Grafana** failures and restored ~100 stuck jobs.

Medisystem GenAI Assisted Fax Classification

- Engineered **Cursor** + **Gemini** (GenAI Gateway) workflows on GCP to classify fax and digipen documents for LTC pharmacy (~2.5M orders/year), with embeddings duplicate detection and a Document Viewer so staff sort by location/home/type for MVP1.

Site Reliability Engineer 2

Loblaw Companies Limited

TORONTO, CANADA

11/2022 – 03/2025

Web UI to manage and trigger SDM Deployments provisioned using Terraform

- Developed a **Flask** web UI (**HTML, Bootstrap, JavaScript**) with a **CloudSQL MySQL** backend and **Azure AD** to manage deployment configs and trigger **GitLab** deployments; deployed on **GKE** with internal DNS and **Istio**.

Google Kubernetes Engine runners for 1000+ Store Deployment Framework and event driven monitoring provisioned using Terraform

- Provisioned a **GKE** cluster of **GitLab** runners for 1000+ store **Helm** deployments (~20k jobs per run, a few times a week) with auto-retry under 1% failure; migrated **Prometheus** exporters to **CloudSQL** (2x performance), converted **Grafana** PROMQL to SQL, and pushed **GitLab** events to **GCP Pub/Sub** via **Cloud Functions** into **MySQL**.

Custom Ansible Framework Development

- Developed a **Rundeck**-style **Ansible** framework to configure SDM stores remotely and modernized the legacy shell deployment onto **GCP** and **Ansible**; migrated 10+ processes (file processing, **IBM MQ**, microservices) and ran 5+ pilots and 10+ rollouts of up to 1200 stores in a single night.

Site Reliability Engineer 1

Loblaw Companies Limited

TORONTO, CANADA

05/2021 – 10/2022

GCP Infra and CI/CD to deploy 20+ apps on Google Kubernetes Engine provisioned using Terraform

- Created a **GitLab** CI/CD catalog for 20+ apps deploying **Docker** artifacts from **GAR** to **GKE** with **Helm**, plus **Terraform** pipelines for **GCP** infra; added **Workload Identity**, **Istio**, and **Secret Manager**, and documented the setup in **Confluence** so devs could run it without SRE.

Dynamic Gitlab Pipelines Infra and Monitoring to support 1000+ store deployment

- Built dynamic nested **GitLab** pipelines (Python, **Jsonnet**) for concurrent deployments to 1000+ SDM stores, with **MIG** runners for 5000+ parallel jobs and a Cloud Function for dynamic runner tags; added auto-scaling **Prometheus** exporters on **GKE** with **Grafana** dashboards and a controller pod to prevent stale/duplicate data.

Personal Projects

TORONTO, CANADA

09/2024 – 10/2025

RenderMuseAI – Text-to-Explainer Animation Platform

Built and launched **RenderMuseAI**, an AI-powered platform that converts text prompts into educational **Manim** animations.

- Used **GitHub Copilot** & **Claude** to generate concept-aware **Manim** code with supervised sub-agents
- Features: **Lovable.dev** frontend, **Supabase** backend, **Daytona** containerized hosting, **Stripe** payments, **Google auth**
- Demo: Fourier Transform explainer [text to explainer animation platform - RenderMuseAI Demo](#)

AI Video Reel Generator

Built a tool to create 1-minute reels from text prompts.

- Integrated **Gemini** for script generation, **Imagen** for images, **GPT-4o-mini** TTS for narration, **FFmpeg** for video + subtitles.
- Frontend: **Lovable.dev**; Backend: **GCP Firebase** + **Cloud Run**.

AI Meal Planning & Recipe App

Built a conversational app for meal planning and grocery management.

- **Flutter** frontend mobile app + **GCP Firebase** backend; **GPT-4o** for grocery bill image scanning & recipe generation.
- **Gemini** multimodal voice-to-voice for recipe CRUD via natural conversation

Skills

AI: Cursor, Gemini, GenAI Gateway, OpenAI, GPT-4o, GPT-4o-mini, Claude, GitHub Copilot, Imagen, embeddings, MCP, Agent Skills, Manim, TTS

Cloud & SRE: GCP, GKE, Google Kubernetes Engine, Kubernetes, Anthos, KubeVirt, FluxCD, GitOps, VMware, Terraform, Ansible, Docker, Helm, Istio, Workload Identity, Secret Manager, GAR, Linux, VPC, DNS, IBM MQ, ActiveMQ, DB2, Postgres, AWS

CI/CD & Observability: GitLab, CI/CD, Jsonnet, MIG, Prometheus, Grafana, Cloud Functions, Pub/Sub, CloudSQL, MySQL, Azure AD, Confluence

Apps & Security: Python, JavaScript, Flask, React, Flutter, HTML, Bootstrap, Firebase, Cloud Run, Supabase, Stripe, FFmpeg, Lovable.dev, Vercel, Daytona, Google Auth, OAuth, OWASP, HL7 FHIR

Certificates

06/2024

Machine Learning Modeling Pipelines in Production Coursera

<https://coursera.org/share/877a1fc7ab16f3d029b9778b98aa41bd>

Machine Learning Data Lifecycle in Production Coursera

<https://coursera.org/share/447273530eeaa5e24b5166f8a383e6da>

12/2020 - 12/2023

AWS (Amazon Web Services) Certified Solutions Architect Associate

https://www.youracclaim.com/badges/1d35a3c5-92664a6c-8848-902f0f05c4a1/public_url

Education

Concordia University

Electrical and Computer Engineering | M.Eng

MONTREAL, CANADA

01/2019 – 12/2020

Graduated with 3.75/4 GPA

Jawaharlal Nehru Technological University

Electronics and Comm Engineering | B.Tech

HYDERABAD, INDIA

09/2014 – 05/2018

SOFT SKILLS: Communication, Team Player, Quick Learner, Innovative.

LANGUAGES: English, Hindi, Telugu.

HOBBIES: Reading, Playing and recording Guitar and Ice Skating